

Gauri Sharma

gsharma80@gatech.edu | +1 (470) 558-7246 | [linkedin.com/in/gs-softwaredev](https://www.linkedin.com/in/gs-softwaredev) | [gauri-sharmaa.github.io](https://github.com/gauri-sharmaa) | github.com/gauri-sharmaa

EDUCATION

Georgia Institute of Technology	2024 – 2028
<i>B.S./M.S. in Computer Science (Integrated Accelerated Program)</i>	GPA: 3.87/4.0
Relevant Coursework: Data Structures & Algorithms, Intro to AI, Computer Organization, Linear Algebra, Discrete Math Certifications: Stanford ML Specialization, AWS Certified Cloud Practitioner, KCNA, Bloomberg Market Concepts	

PUBLICATIONS

Echoes of Human Malice in Agents: Benchmarking LLMs for Multi-Turn Online Harassment Attacks. ICWSM 2027.

PROFESSIONAL EXPERIENCE

Software Development Engineer Intern	May 2026 – August 2026
<i>Amazon Web Services</i>	Seattle, WA
- Built a production Java microservice powering compliance review workflows for 500+ internal users globally, owning Smithy APIs, CDK infrastructure, DynamoDB persistence, SNS eventing, and TestNG integration tests across 6 service packages. - Designed a compensating-transaction write path across S3, two DynamoDB tables, and SNS to guarantee consistency. - Implemented end-to-end request idempotency with composite keys, conflict detection, and conditional-write race handling.	
Research Assistant	Jan 2025 – Present
<i>GT Financial Services Innovation Lab</i>	Atlanta, GA
- Built a macro forecasting pipeline (ADSN shapelets, Granger causality, RidgeCV) for jobless claims and fed funds. - Evaluated forecasts with RMSE and Diebold-Mariano tests, improving accuracy by 12% over autoregressive (AR) baselines. - Built compliance retrieval benchmarks (FATF standards, consumer-finance policy, card disclosures) for embedding models.	
Software Engineering Intern	May 2025 – July 2025
<i>IBM Research for Open Source</i>	Atlanta, GA
- Built a cross-cluster krew plug-in in Golang with 46 custom kubectl commands for KubeStellar, a CNCF Kubernetes project. - Parallelized per-cluster Kubernetes API queries with goroutines, cutting cross-cluster workload aggregation latency by 72%. - Queried resources with client-go's dynamic client across kubeconfig contexts; covered by unit and e2e tests on kind clusters.	
Agentic Cybersecurity Fellow	May 2025 – Aug 2025
<i>Fusen World Fellowship</i>	Atlanta, GA
- Co-founded AegentDev, an AI agent visibility and security platform; granted \$30K in GCP credits by Google. - Built an agentic security harness that connects to CrewAI and LangGraph agents over a Model Context Protocol (MCP) interface, runs agent-to-agent security scans in Docker-based deployments, and cut end-to-end scan time by 32%. - Prototyped an opinionated Agent Cascade Injection (ACI) attack framework with MIT NANDA (arXiv:2507.21146). - Invited to YC Startup School (2026) to keep building AegentDev with a co-founder after the fellowship ended.	
Research Assistant	Jan 2025 – Present
<i>GT Social Dynamics & Wellbeing Lab</i>	Atlanta, GA
- Co-authored a study benchmarking jailbroken LLM agents in multi-turn online harassment, accepted to ICWSM 2027. - Built an open-source image PII detector with YOLO, CLIP, and a VLM, at 73% accuracy on VISPR categories. - Replicating FiG-Priv (COLM 2025), a multi-agent VLM privacy-masking pipeline, as a baseline for the image PII detector.	

PROJECTS

Millennium-Supervised Low-Latency Trading Systems Project <i>Go, FIX 4.2</i>	Jan 2026 – May 2026
- Built a low-latency order execution engine in Go with a 3-person team, using a lock-free ring buffer, WAL recovery, FIX 4.2 connectivity, and inline risk checks to achieve 26μs engine latency; presented the system at Millennium's Miami office. - Designed the engine's idempotent order API with request keys, dedup state, and safe retries after client-side timeouts. - Built a zero-allocation ML momentum predictor that runs inline on the order path with approximately 100ns inference.	
LLM Routing Research <i>Python, PyTorch, Hugging Face, HPC</i>	Jan 2026 – Present
- Researching cost-aware routing that sends each prompt to the cheapest capable LLM. - Showed grouping prompts by difficulty beats topic clustering with under 500 labeled prompts (+0.011 to +0.017 area). - Found headroom is anti-correlated with oracle gain (Spearman -0.50, 27 tasks), a model-pool confound in router evaluation. - Awarded a \$1K OpenAI Research Grant and Georgia Tech's President's Undergraduate Research Award (PURA).	
Inference-Time Steering for Frozen VLA Policies (ML 7641) <i>PyTorch, XGBoost, OpenVLA, LIBERO</i>	2026
- Led the constraint-violation evaluator, auto-labeling 6,405 windows with zero human annotation to train a steering critic. - Showed constraint semantics form separable clusters in OpenVLA's frozen Llama-2 encoder across 630 paraphrased instructions.	

LEADERSHIP

Currencies Team Lead, Market Insights Head <i>Trading @ Georgia Tech</i>	Jan 2025 – Present
Led weekly FX research, turning central bank signals into investment briefs for 400+ student investors; mentored 6 analysts.	
Partnerships Lead <i>BDBI @ Georgia Tech</i>	Jan 2025 – Present
Led sponsorships and external partnerships for Big Data Big Impact (BDBI), supporting 30+ events over 3 years through industry collaborations, technical workshops, mentorship programs, recruiting initiatives, and partnerships with companies and startup organizations across Georgia Tech.	

SKILLS

Languages: Bash/Zsh, C, C++, Go (Golang), HTML/CSS, Java, JavaScript, Python, R, SQL, TypeScript, Assembly
Libraries/Frameworks: Flask, Hugging Face, LangChain, MCP, NumPy, Pandas, PyTorch, React, scikit-learn, TensorFlow
Tools: AWS, CI/CD, Docker, Firebase, GCP, Git, Jira, Jupyter, Kubernetes, MySQL, REST APIs, Snowflake, SQLite, Tableau
Other: Agile, Cloud Computing, Computer Vision, Deep Learning, GenAI, Machine Learning, Multimodal Sentiment Analysis, NLP